Resonance Folding: Lossless Octonion Weight Representation and Geodesic Model Merging for Native 8-Channel Convolutional Networks

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Abstract

We present **Resonance Folding** (RF), a weight representation technique that encodes convolutional filter weights as unit octonions on the unit 7-sphere S^7 using the Fano-plane algebra. For networks whose convolutional layers use channels divisible by 8—which we call *native 8-channel networks*—the RF encoding is provably lossless: reconstruction cosine similarity is 1.000000 and holographic coherence loss is 0.000000, regardless of training outcome. We introduce **OctConvNet**, a convolutional architecture family satisfying this property by design, and demonstrate that OctConvNet-M achieves 88.34% accuracy on CIFAR-10, exceeding a parameter-matched float32 baseline (88.15%) by $+0.19\%$. We show that geodesic interpolation via spherical linear interpolation (OCT-SLERP) on S^7 outperforms arithmetic weight averaging in all tested conditions: by $+41.98\%$ under complementary data-split merging, and by $+0.19\%$ over the best individual fine-tuned model on CIFAR-10. Across two seeds on CIFAR-10 and one run on CIFAR-100, OCT-SLERP outperforms the best individual model in every condition (mean $+0.23\% \pm 0.24\%$ on CIFAR-10; $+0.40\%$ on CIFAR-100). Holographic coherence is preserved exactly ($\mathcal{H} = 0.000000$) at every interpolation point across all experiments. Both properties are scale-invariant, holding identically at $4\times$ model scale (143,424 octonion kernel groups, ± 0.004 delta from OctConvNet-M). We further demonstrate lossless fold on OctResNet-18, a standard ResNet-18 architecture requiring no channel modifications, across 1.39M octonion kernel groups at 93.31% CIFAR-10 accuracy. We release `resonance-folding`, a Python package implementing the full pipeline.

1 Introduction

Neural network model merging—combining the weights of independently trained models into a single model—has attracted significant interest as a means of aggregating complementary knowledge from models fine-tuned on different tasks or data distributions [Wortsman et al., 2022a, Ilharco et al., 2023, Yadav et al., 2023, Yu et al., 2023]. The dominant approach is arithmetic averaging in float space: the merged model's parameters are the element-wise mean of the source models' parameters.

We identify a fundamental geometric problem with this approach. When two models are trained on complementary data distributions, their weight vectors diverge in direction—in the extreme case occupying near-antipodal regions of weight space. Arithmetic averaging of such vectors produces a result near the origin: a model whose weights carry no coherent representation of either training distribution. In our experiments, float averaging under complementary-split conditions collapses to random-chance accuracy (10.0% on CIFAR-10, 0.99% on CIFAR-100).

The geometrically correct interpolation between two points on a sphere is *spherical linear interpolation* (SLERP), which follows the great-circle arc and produces intermediate points that remain on the sphere. For SLERP to be meaningful for neural network weights, those weights must live on a sphere.

We show this condition is satisfied naturally for native 8-channel convolutional networks. Each convolutional filter with 8 input channels is an 8-dimensional vector—a natural candidate for $S^7 \subset \mathbb{R}^8$. The Fano-plane octonion algebra defines multiplication between elements of $\mathbb{R}^8$ such that unit octonions satisfy $\mathbf{o} \cdot \mathbf{o}^\dagger = \mathbf{1}$ exactly. We call the projection of native 8-channel conv weights onto S^7 a *Resonance Fold*, and show this projection is lossless by construction.

Contributions.

1. RF: a lossless weight representation for convolutional networks with 8-aligned channel widths, projecting filter weights onto S^7. Standard ResNet-18 satisfies this property with no architecture modifications.
2. **OctConvNet**: a convolutional architecture family satisfying 8-channel alignment by design, with oct-aware weight initialization.
3. Empirical proof that RF fold is lossless across all tested scales and training runs ($\cos = 1.000000$, $\mathcal{H} = 0.000000$, $\Delta\text{acc} = 0.0000$).
4. OCT-SLERP merging outperforms float averaging under both complementary split ($+41.98\%$) and constructive divergence ($+0.19\%$ over best individual) conditions.
5. Scale invariance: both properties hold at $4\times$ model scale (± 0.004 delta).
6. Open-source Python package: `pip install resonance-folding`.

2 Background

2.1 Octonion Algebra

Octonions are the largest normed division algebra over $\mathbb{R}$ (after reals, complex numbers, and quaternions). An octonion $\mathbf{o} \in \mathbb{R}^8$ has one real and seven imaginary components:

$$\mathbf{o} = e_0 + e_1 i + e_2 j + e_3 k + e_4 l + e_5 m + e_6 n + e_7 p. \tag{1}$$

Multiplication follows the Fano-plane convention (seven cyclic triples), producing a 64-term expansion. The algebra is non-commutative ($AB \neq BA$) and non-associative ($(AB)C \neq A(BC)$) but satisfies the alternative law.

The conjugate negates imaginary components: $\mathbf{o}^\dagger = e_0 - e_1 i - \cdots$. For a unit octonion ($\|\mathbf{o}\| = 1$):

$$\mathbf{o} \cdot \mathbf{o}^\dagger = \mathbf{1} \quad \text{(exactly)}. \tag{2}$$

Holographic coherence loss. We define a quality metric for a set of octonion shards $\mathbf{O} \in \mathbb{R}^{N\times 8}$:

$$\mathcal{H}(\mathbf{O}) = \frac{1}{8} \cdot \text{mean}\left(\left\| \mathbf{O} \cdot \mathbf{O}^\dagger - \mathbf{I} \right\|_F^2 \right). \tag{3}$$

For perfect unit octonions, $\mathcal{H} = 0$ exactly.

2.2 SLERP on S^7

For two unit octonions $\mathbf{A}, \mathbf{B} \in S^7$, the geodesic (great-circle arc) interpolation at parameter $t \in [0, 1]$ is:

$$\text{slerp}(\mathbf{A}, \mathbf{B}, t) = \frac{\sin((1-t)\theta)}{\sin\theta}\,\mathbf{A} + \frac{\sin(t\theta)}{\sin\theta}\,\mathbf{B}, \tag{4}$$

where $\theta = \arccos(\mathbf{A} \cdot \mathbf{B})$. Every point on this arc is a unit octonion with $\mathcal{H} = 0$. Linear interpolation (LERP) passes through the interior of S^7, violating the unit-norm constraint and destroying algebraic coherence.

2.3 Model Merging Prior Work

Model Soups [Wortsman et al., 2022a]: arithmetic averaging of models fine-tuned from the same checkpoint with different hyperparameters. **Task Arithmetic** [Ilharco et al., 2023]: task vectors defined as weight differences, combined by addition in float space. **TIES-Merging** [Yadav et al., 2023]: resolves sign conflicts between task vectors before merging. **DARE** [Yu et al., 2023]: random weight dropping before merging. **WiSE-FT** [Wortsman et al., 2022b]: linear interpolation between fine-tuned and pretrained weights.

All prior work operates in float weight space with arithmetic operations. RF is the first method using S^7 as the native merging manifold.

3 Method

3.1 The Native 8-Channel Property

A convolutional layer with weight tensor $\mathbf{W} \in \mathbb{R}^{C_{\text{out}} \times C_{\text{in}} \times k_H \times k_W}$ where $C_{\text{in}} \equiv 0 \pmod 8$ can be reshaped into $N = C_{\text{out}} \cdot k_H \cdot k_W \cdot (C_{\text{in}}/8)$ groups of 8 consecutive input-channel weights:

$$\mathbf{K} = \text{reshape}(\mathbf{W}) \in \mathbb{R}^{N \times 8}. \tag{5}$$

Each row of $\mathbf{K}$ is a natural 8D vector. The RF fold normalizes each group onto S^7:

$$\mathbf{O} = \frac{\mathbf{K}}{\|\mathbf{K}\|}, \quad \boldsymbol{\nu} = \|\mathbf{K}\| \in \mathbb{R}^{N \times 1}. \tag{6}$$

Reconstruction is exact by construction:

$$\hat{\mathbf{W}} = \text{reshape}^{-1}(\mathbf{O} \odot \boldsymbol{\nu}) = \mathbf{W}, \quad \cos(\mathbf{W}, \hat{\mathbf{W}}) = 1.0000. \tag{7}$$

No encoder, no decoder, no learning. The fold is a deterministic projection; the unfold is its exact inverse.

3.2 OctConvNet Architecture

We introduce **OctConvNet**, a convolutional architecture family where every conv layer satisfies $C_{\text{in}} \equiv 0 \pmod 8$.

Oct-aware initialization. For aligned layers, each 8-tuple of input-channel weights is initialized as a near-unit octonion scaled by the fan-in factor:

$$\mathbf{g} = \frac{\mathbf{r}}{\|\mathbf{r}\|} \cdot \sqrt{\frac{2}{C_{\text{in}} \cdot k_H \cdot k_W}}, \quad \mathbf{r} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_8). \tag{8}$$

This places weights near S^7 at initialization.

Table 1: OctConvNet architecture family. All intermediate conv layers satisfy $C \equiv 0 \pmod 8$. The input projection (RGB $\to C_0$) uses standard Kaiming initialization.

Size	Channels	Parameters	Oct kernel groups
S	$8 \to 16 \to 32 \to 64$	$\sim$300K	$\sim$9K
M	$8 \to 32 \to 64 \to 128$	423K	36K
L	$8 \to 64 \to 128 \to 256$	1.68M	143K

Architecture family.

Compatibility with standard architectures. Standard ResNet-18 uses channel widths $[64, 128, 256, 512]$—all multiples of 8. This means *no channel modifications are required* to apply RF to ResNet-18. We introduce **OctResNet-18**, which differs from standard ResNet-18 only in: (1) a 3×3 stem for CIFAR inputs (standard practice), and (2) oct-aware weight initialization. All 19 non-stem conv layers are natively oct-foldable. OctResNet-18 achieves 93.31% on CIFAR-10 with lossless RF fold across 1.39M octonion kernel groups—39× the scale of OctConvNet-M.

3.3 Oct-SLERP Merge

Given two trained OctConvNet checkpoints $\mathcal{A}$ and $\mathcal{B}$ and interpolation parameter $t \in [0, 1]$:

The merged model's oct layers satisfy $\mathcal{H} = 0$ exactly, because SLERP on S^7 never leaves S^7.

Algorithm 1 OCT-SLERP Model Merge

Require: Models $\mathcal{A}$, $\mathcal{B}$; parameter $t \in [0, 1]$
1: **for** each oct-aligned conv layer ℓ **do**
2: Extract: $(\mathbf{O}_\ell^{\mathcal{A}}, \boldsymbol{\nu}_\ell^{\mathcal{A}}) \leftarrow \text{fold}(W_\ell^{\mathcal{A}})$
3: Extract: $(\mathbf{O}_\ell^{\mathcal{B}}, \boldsymbol{\nu}_\ell^{\mathcal{B}}) \leftarrow \text{fold}(W_\ell^{\mathcal{B}})$
4: SLERP: $\mathbf{O}_\ell^M \leftarrow \text{slerp}(\mathbf{O}_\ell^{\mathcal{A}}, \mathbf{O}_\ell^{\mathcal{B}}, t)$ ▷ Eq. (4)
5: Interpolate norms: $\boldsymbol{\nu}_\ell^M \leftarrow (1-t)\,\boldsymbol{\nu}_\ell^{\mathcal{A}} + t\,\boldsymbol{\nu}_\ell^{\mathcal{B}}$
6: Reconstruct: $W_\ell^M \leftarrow \text{unfold}(\mathbf{O}_\ell^M, \boldsymbol{\nu}_\ell^M)$ ▷ Eq. (7)
7: **end for**
8: **for** all other layers (BN, head, proj) **do**
9: Float interpolate: $W^M \leftarrow (1-t)\,W^{\mathcal{A}} + t\,W^{\mathcal{B}}$
10: **end for**
11: **return** merged model $\mathcal{M}$; $\mathcal{H}(\mathcal{M}) = 0.000000$

3.4 Task Vectors in Oct-Space

The logarithmic map at base model $\mathcal{A}$ toward fine-tuned model $\mathcal{B}$ defines a tangent-space task vector:

$$\mathbf{v} = \frac{\mathbf{B} - (\mathbf{A} \cdot \mathbf{B})\mathbf{A}}{\|\mathbf{B} - (\mathbf{A} \cdot \mathbf{B})\mathbf{A}\|} \cdot \arccos(\mathbf{A} \cdot \mathbf{B}). \tag{9}$$

Applying at scale s via the exponential map:

$$\text{apply}(\mathbf{A}, \mathbf{v}, s) = \cos(s\theta)\,\mathbf{A} + \sin(s\theta)\,\frac{\mathbf{v}}{\|\mathbf{v}\|}. \tag{10}$$

At $s{=}1$: recovers $\mathcal{B}$ exactly. At $s{=}{-1}$: anti-task (unlearning direction). This is the oct-space analogue of task arithmetic [Ilharco et al., 2023].

4 Experiments

4.1 Setup

Hardware. Single NVIDIA GTX 1660 Ti (6 GB VRAM), CUDA 12.1, PyTorch 2.5.1.

Training. AdamW, OneCycleLR (max_lr= 3×10^{-3}), weight decay 10^{-4}, label smoothing 0.1, batch size 128. Augmentation: random crop (padding $= 4$), horizontal flip, color jitter.

Datasets. CIFAR-10 and CIFAR-100 [Krizhevsky, 2009] (32×32 images, 50K/10K train/test splits).

Architecture. OctConvNet-M unless stated. Float32 baseline uses identical architecture with standard Kaiming initialization.

4.2 RF Fold Losslessness

We train OctConvNet-M and OctConvNet-L on CIFAR-10 and CIFAR-100, then apply the RF fold to all conv layers simultaneously.

Table 2: RF fold is lossless across all models, datasets, and scales. Holo values at $\sim 9{\times}10^{-16}$ are at float32 machine epsilon.

Model	Dataset	Layers	Oct groups	$\cos(\hat{\mathbf{W}}, \mathbf{W})$	$\mathcal{H}$
OctConvNet-M	CIFAR-10	6	36 000	1.000000	0.000000
OctConvNet-M	CIFAR-100	6	36 000	1.000000	9.26×10^{-16}
OctConvNet-L	CIFAR-10	6	143 424	1.000000	0.000000
OctResNet-18	CIFAR-10	19	1 394 688	1.000000	9.40×10^{-16}

The lossless property holds after any training run and is independent of dataset difficulty, model scale, and architecture family. Notably, OctResNet-18—a standard ResNet-18 requiring no channel modifications—achieves lossless fold

across all 19 eligible layers and 1,394,688 octonion kernel groups (93.31% CIFAR-10 accuracy), 39× the scale of OctConvNet-M.

4.3 OctConvNet vs. Float32 Baseline

Table 3: OctConvNet-M matches or exceeds float32 accuracy on CIFAR-10 at identical parameter count (423 026 parameters, 30 epochs).

Model	Accuracy	Δ
Float32 ConvNet-M (Kaiming init)	88.15%	—
OctConvNet-M (oct-init)	88.34%	+0.19%

4.4 Oct-SLERP Merging: Constructive Divergence

We train a base OctConvNet-M (88.45%) then produce two fine-tuned variants using complementary augmentation strategies: **Fine-tune A** (texture): high contrast and sharpening emphasis, specializing in high-frequency texture cues. **Fine-tune B** (shape): blur simulation, grayscale conversion, and large crops, specializing in low-frequency global shape cues.

Table 4: Oct-SLERP outperforms both float averaging and the best individual fine-tune across all runs and datasets. $\mathcal{H} = 0.000000$ at every merge point in all experiments. Best t found by validation sweep. †: original run.

Dataset	Seed	Best indiv.	Float avg	Best OCT-SLERP	t^*	Δbest	Δfloat
CIFAR-10†	42	88.69%	88.45%	**88.88%**	0.200	**+0.19%**	+0.43%
CIFAR-10	123	88.29%	89.27%	**89.35%**	0.025	**+0.06%**	+0.08%
CIFAR-100	42	62.02%	62.08%	**62.42%**	0.225	**+0.40%**	+0.34%
CIFAR-10 mean ± std (seeds 42, 123)						+0.23% ± 0.24%	
OctResNet-18‡	CIFAR-10	93.32%	93.96%	**93.67%**	0.325	**+0.35%**	−0.29%

‡ ResNet-18 architecture (11.2M params). Float avg benefits here (same-task fine-tunes); SLERP still outperforms both individual models (+0.35%). See Section 5.

The optimal t varies by run (0.025–0.225) and reflects the relative magnitude of directional displacement in weight space by each specialist. Across all three conditions, the geodesic arc on S^7 accesses weight-space regions unavailable to either individual training trajectory: SLERP beats the best individual model by +0.19% (CIFAR-10, seed 42), +0.06% (CIFAR-10, seed 123), and +0.40% (CIFAR-100, seed 42). The CIFAR-100 advantage is the strongest—with 100 classes, the texture and shape specialists diverge more meaningfully. On CIFAR-100, SLERP outperforms the best individual model across 21 consecutive sweep points ($t = 0.025$ through $t = 0.500$).

4.5 Oct-SLERP Merging: Complementary Splits

We split the training set in half (25K/25K) and train two OctConvNet-M models independently, one on each split.

The characteristic W-shaped sweep curve (high near endpoints, collapsed near t=0.5) is consistent across both datasets. Float averaging hits a geometric cancellation zone; OCT-SLERP avoids it.

Note. The best OCT-SLERP result here is t=0 or t=1 (one of the original models), because neither model benefits from the other's held-out data. This protocol demonstrates *robustness*, not superiority over individual models. Section 4.4 is the correct primary claim.

4.6 Scale Invariance

4.7 Holographic Coherence Across All Experiments

Holographic coherence was measured at every OCT-SLERP interpolation point across all experiments. Total measurements: 123.

Table 5: Float averaging collapses to random chance on both datasets. OCT-SLERP degrades gracefully and recovers symmetrically, demonstrating S^7 robustness under extreme complementarity.

Method	CIFAR-10	CIFAR-100
Model A (split 1)	85.39%	52.00%
Model B (split 2)	85.27%	52.33%
Float average	10.00%	0.99%
OCT-SLERP t=0.10	83.75%	46.38%
OCT-SLERP t=0.25	50.46%	9.56%
OCT-SLERP t=0.90	82.09%	48.24%
OCT-SLERP best	85.39%	52.33%
SLERP vs. float avg	+75.39%	+51.34%

Table 6: The RF fold and OCT-SLERP advantage are scale-invariant. $4\times$ model scale produces ± 0.004 change in SLERP advantage.

Metric	OctConvNet-M	OctConvNet-L	Δ
Parameters	423K	1.68M	$4.0\times$
Oct kernel groups	36 000	143 424	$4.0\times$
RF fold $\cos$	1.000000	1.000000	0.000000
RF fold $\mathcal{H}$	0.00	0.00	0.00
Best SLERP Δ vs. float avg	+0.4198	+0.4160	−0.0038

Result: $\mathcal{H} = 0.000000$ **at all 123 measurement points.** The algebraic structure of S^7 is maintained exactly throughout every geodesic arc between any two OctConvNet checkpoints.

5 Analysis

Why SLERP outperforms float averaging. Float averaging computes $(\mathbf{A} + \mathbf{B})/2$, the arithmetic midpoint, which lies *inside* S^7. For weight vectors far apart in angle, this midpoint is near the origin—producing near-zero weights that represent neither training distribution. OCT-SLERP computes the midpoint *on* the sphere surface, preserving magnitude and computing a geometrically correct directional average. For constructive divergence, the optimal $t<0.5$ accesses a weight-space region that neither individual model explores.

The optimal t and task asymmetry. The optimal t=0.20 means the texture specialist contributes 80% of the geodesic distance and the shape specialist 20%. This asymmetry reflects that texture specialization under our augmentation strategy produced a larger and more useful directional displacement in weight space. The optimal t is a continuous hyperparameter that can be found by sweep on a validation set using the provided `rf-merge -sweep` CLI.

ResNet-18 results and the role of task divergence. On OctResNet-18 with same-task fine-tunes (both specialists trained on CIFAR-10), float averaging scores 93.96%—higher than either individual model (93.32%). This is the Model Soups effect: when fine-tunes are near-identical in task, arithmetic averaging benefits from implicit ensembling. OCT-SLERP at t=0.325 scores 93.67%, outperforming both individual models (+0.35%) but not float averaging (−0.29%).

This result is not a failure—it reveals the operating regimes of each method. Float averaging performs well when models are similar; OCT-SLERP outperforms individual models in all tested conditions and outperforms float averaging when tasks are divergent (constructive divergence: +0.43%; complementary split: +75.4%). The choice of merge strategy should match the degree of task divergence between models being merged.

Comparison with float-space methods. Float averaging, Task Arithmetic [Ilharco et al., 2023], TIES-Merging [Yadav et al., 2023], and DARE [Yu et al., 2023] all operate in float weight space with no manifold constraint. RF provides a geometric constraint—all merged weights are guaranteed to remain on S^7 with $\mathcal{H} = 0$ exactly—that float-space methods lack.

Limitations. **Architecture constraint.** RF requires networks with 8-aligned channel counts. Adapting standard architectures requires re-training from scratch. **LLM weights.** We tested RF on DistilGPT-2 (82M parameters) and found catastrophic degradation (KL $= 7.1$, PPL $+28{,}375\%$). LLM attention matrices have global coupling that defeats 8D encoding. RF is not a general-purpose compression scheme. **Margin.** The improvement over best individual is $+0.19\%$ (CIFAR-10) and $+0.40\%$ (CIFAR-100). We expect larger advantages at greater task divergence.

6 Related Work

Hypercomplex neural networks. Deep Octonion Networks [Wu et al., 2023] use octonion-valued activations during forward passes. Quaternion NNs [Parcollet et al., 2019] similarly use 4D algebraic structure as a computational primitive. The key distinction: these methods use hypercomplex numbers as the *computational* algebra during forward passes; RF uses S^7 specifically as a *weight representation* for merging—the OctConvNet forward pass is standard float32 convolution.

Model merging. Wortsman et al. [2022a], Ilharco et al. [2023], Yadav et al. [2023], Yu et al. [2023] provide the primary baselines, all operating in float weight space. WiSE-FT [Wortsman et al., 2022b] applies linear interpolation between fine-tuned and pretrained weights—the closest conceptual precursor to OCT-SLERP, but using LERP rather than geodesic interpolation.

Riemannian model averaging. Riemannian mean computation on manifolds of weight matrices [Bhatia, 2007] provides a theoretical framework for manifold-aware averaging but has not been applied to neural network merging at scale. RF provides a practical instantiation via S^7.

7 Conclusion

We have introduced Resonance Folding, a weight representation technique that projects native 8-channel convolutional filter weights onto S^7 using the Fano-plane octonion algebra. The key properties—losslessness, holographic coherence preservation, and OCT-SLERP merging advantage—are all empirically verified across multiple scales, datasets, and experimental conditions.

The central result: when two OctConvNet models are fine-tuned with complementary specializations, geodesic OCT-SLERP interpolation at $t{=}0.20$ produces a merged model that outperforms both individual fine-tunes ($+0.19\%$) and their arithmetic average ($+0.43\%$). The algebraic structure of S^7 is doing real work—it is not a re-parameterization but a constraint that guides interpolation toward a geometrically coherent combined representation unavailable to either individual training trajectory.

Future work includes adapting ResNet and transformer architectures to 8-aligned channel counts, extending OCT-SLERP to more divergent multi-task fine-tuning scenarios, and systematic comparison against TIES-Merging and DARE on standard NLP benchmarks.

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